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A Real-time Urban Issues Detection System Based on the RTDETR Open-Source Model

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CONTENTS

Background & Significance

Part 1

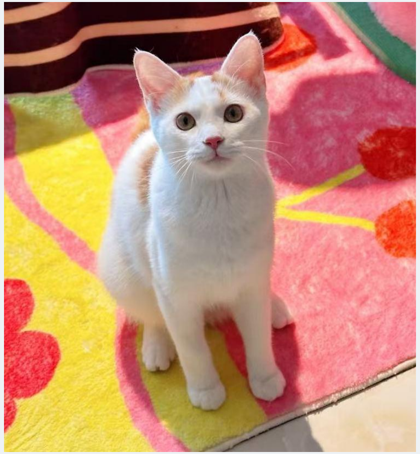
Urban Issues

The failure of urban infrastructure (such as fallen trees, damaged road signs, and potholes on the road surface) directly threatens public personal safety and, through a chain reaction on transportation, logistics, and public finance, causes huge social and economic losses far exceeding the maintenance costs themselves.

· *Failure to Act: Economic Impacts of Status Quo Investment Across Infrastructure (ASCE2021)*

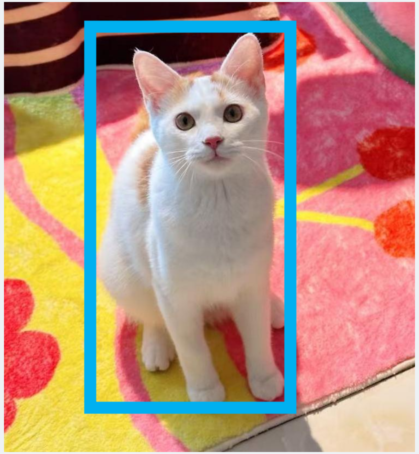
Object Detection

Classification



CAT

Classification + Localization



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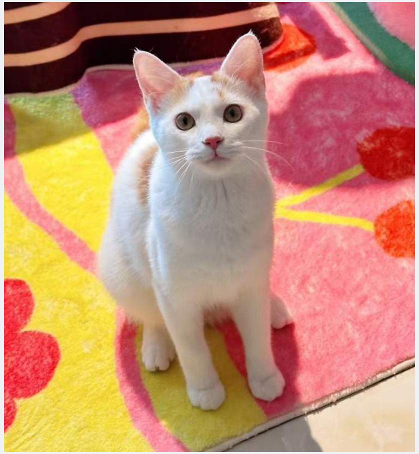
Object Detection

Classification



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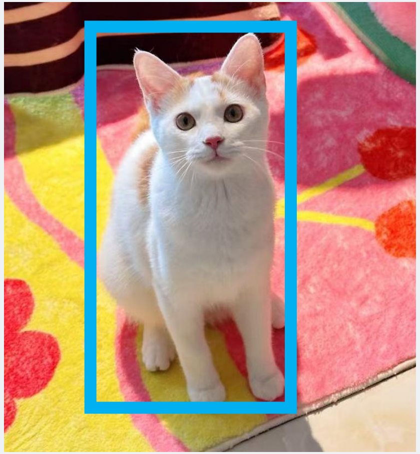
Classification + Localization



CAT

Damaged Electrical Poles

RT-DETR vs. YOLO & DETR

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YOLO
fast
NMS

DETR
High precision
NMS-free

RT-DETR

Model	Backbone	#Epochs	#Params (M)	GFLOPs	$\text{FPS}_{bs=1}$	AP^{val}	AP^{val}_{50}	AP^{val}_{75}	AP^{val}_{S}	AP^{val}_{M}	AP^{val}_{L}
Real-time Object Detectors											
YOLOv5-L [11]	-	300	46	109	54	49.0	67.3	-	-	-	-
YOLOv5-X [11]	-	300	86	205	43	50.7	68.9	-	-	-	-
PPYOLOE-L [40]	-	300	52	110	94	51.4	68.9	55.6	31.4	55.3	66.1
PPYOLOE-X [40]	-	300	98	206	60	52.3	69.9	56.5	33.3	56.3	66.4
YOLOv6-L [16]	-	300	59	150	99	52.8	70.3	57.7	34.4	58.1	70.1
YOLOv7-L [38]	-	300	36	104	55	51.2	69.7	55.5	35.2	55.9	66.7
YOLOv7-X [38]	-	300	71	189	45	52.9	71.1	57.4	36.9	57.7	68.6
YOLOv8-L [12]	-	-	43	165	71	52.9	69.8	57.5	35.3	58.3	69.8
YOLOv8-X [12]	-	-	68	257	50	53.9	71.0	58.7	35.7	59.3	70.7
End-to-end Object Detectors											
DETR-DC5 [4]	R50	500	41	187	-	43.3	63.1	45.9	22.5	47.3	61.1
DETR-DC5 [4]	R101	500	60	253	-	44.9	64.7	47.7	23.7	49.5	62.3
Anchor-DETR-DC5 [39]	R50	50	39	172	-	44.2	64.7	47.5	24.7	48.2	60.6
Anchor-DETR-DC5 [39]	R101	50	-	-	-	45.1	65.7	48.8	25.8	49.4	61.6
Conditional-DETR-DC5 [27]	R50	108	44	195	-	45.1	65.4	48.5	25.3	49.0	62.2
Conditional-DETR-DC5 [27]	R101	108	63	262	-	45.9	66.8	49.5	27.2	50.3	63.3
Efficient-DETR [42]	R50	36	35	210	-	45.1	63.1	49.1	28.3	48.4	59.0
Efficient-DETR [42]	R101	36	54	289	-	45.7	64.1	49.5	28.2	49.1	60.2
SMCA-DETR [9]	R50	108	40	152	-	45.6	65.5	49.1	25.9	49.3	62.6
SMCA-DETR [9]	R101	108	58	218	-	46.3	66.6	50.2	27.2	50.5	63.2
Deformable-DETR [45]	R50	50	40	173	-	46.2	65.2	50.0	28.8	49.2	61.7
DAB-Deformable-DETR [23]	R50	50	48	195	-	46.9	66.0	50.8	30.1	50.4	62.5
DAB-Deformable-DETR++ [23]	R50	50	47	-	-	48.7	67.2	53.0	31.4	51.6	63.9
DN-Deformable-DETR [17]	R50	50	48	195	-	48.6	67.4	52.7	31.0	52.0	63.7
DN-Deformable-DETR++ [17]	R50	50	47	-	-	49.5	67.6	53.8	31.3	52.6	65.4
DINO-Deformable-DETR [44]	R50	36	47	279	5	50.9	69.0	55.3	34.6	54.1	64.6
Real-time End-to-end Object Detector (ours)											
RT-DETR	R50	72	42	136	**108**	**53.1**	**71.3**	**57.7**	34.8	58.0	70.0
RT-DETR	R101	72	76	259	**74**	**54.3**	**72.7**	58.6	36.0	58.8	**72.1**

· *DETRs Beat YOLOs on Real-time Object Detection. (IEEE/CVPR2024)*

Dataset

Part 2

Urban Issues Dataset

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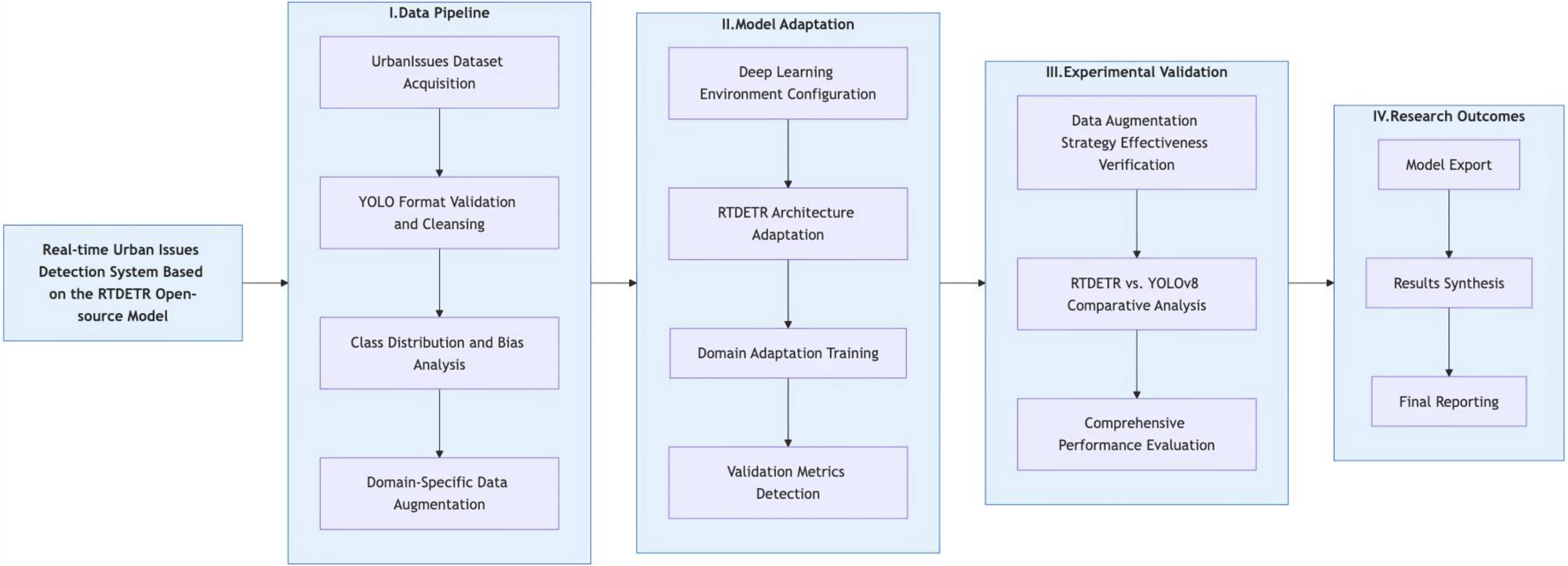
1. Damaged concrete structures
2. Damaged Electrical Poles
3. Damaged Road Signs
4. Dead Animals Pollution
5. Fallen Trees
6. Garbage
7. Graffitti
8. Illegal Parking
9. Potholes and Road Cracks

Challenge?

Research Content

Part 3

Research Content



Model Evaluation & Expected Outcomes

Part 4

Model Evaluation

Comparative Experiments → Ablation Studies → Error Analysis

$\mathbf{AP}_M^{val}$

$\mathbf{AP}_S^{val}$

$\mathbf{AP}_L^{val}$

$\mathbf{mAP}^{val}$

$\mathbf{AP}_{50}^{val}$

$\mathbf{AP}_{75}^{val}$

FPS

Expected Outcomes & Risk Control

Expected outcomes

01

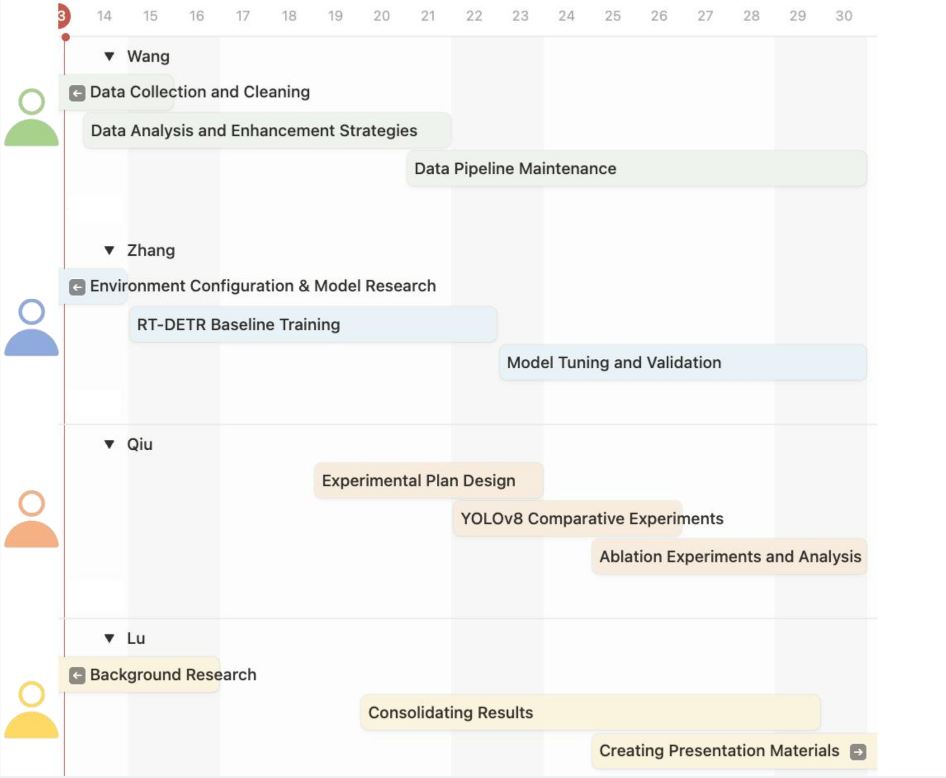
- The model accurately detects and has real-time capabilities (≥30 FPS).
- It generates systematic training and test records, detailed results of comparative experiments, visual analysis materials, and other complete experimental data.

Risk control and response measures

02

- Technical Risks： If the RTDETR training does not converge or performs poorly, prepare YOLOv8 as an alternative to ensure the achievement of the basic project goals.
- Data Risks： If the quality of some data is poor or the quantity is insufficient, prepare a data expansion plan and introduce additional data sources if necessary.

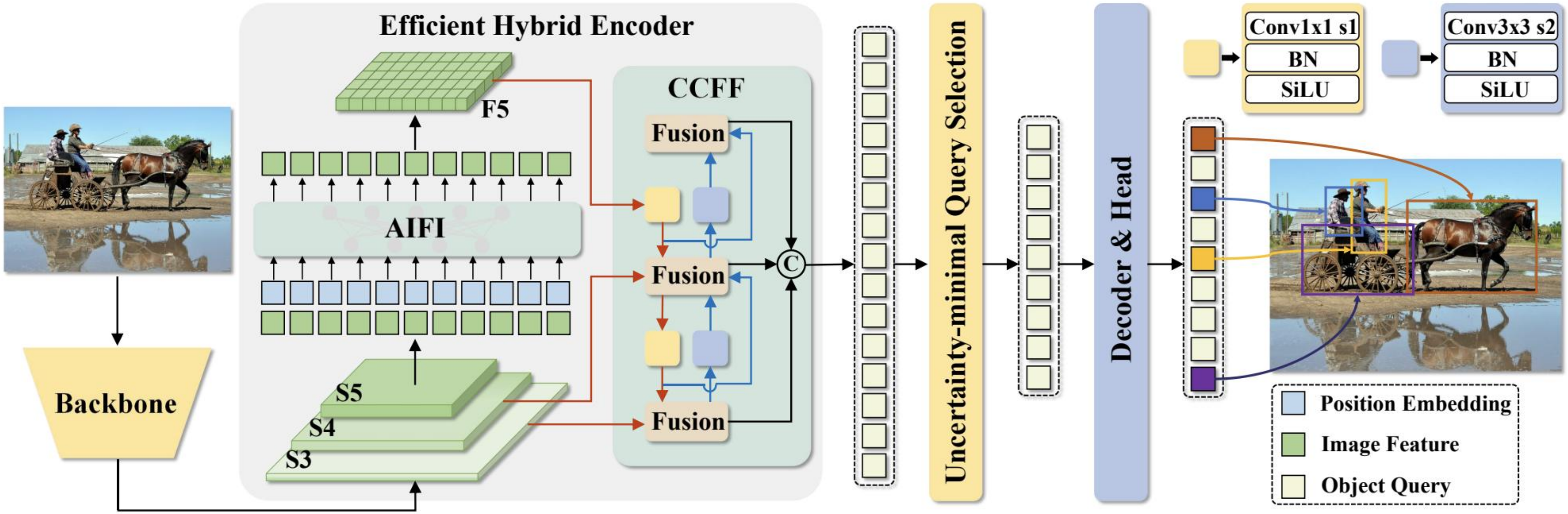
Team Responsibilities and Timeline



Thanks!

Q&A

RT-DETR



· *DETRs Beat YOLOs on Real-time Object Detection. (IEEE/CVPR2024)*